UNITED STATES DISTRICT COURT
FOR THE DISTRICT OF COLUMBIA

UNITED STATES OF AMERICA et al.,	)	
Plaintiffs,	)	
v.	)	**Case No. 20-cv-3010 (APM)**
GOOGLE LLC,	)	
Defendant.	)	
STATE OF COLORADO et al.,	)	
Plaintiffs,	)	
v.	)	**Case No. 20-cv-3715 (APM)**
GOOGLE LLC,	)	
Defendant.	)	

MEMORANDUM OPINION

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INTRODUCTION

The general search engine has revolutionized how we live. Information that once took hours or days to acquire can now be found in an instant on the internet with the help of a general search engine. General search engines use powerful algorithms to create what seems like magic. Enter a search query, and the general search engine will retrieve, rank, and display the websites that provide the exact information the user seeks at that very moment. And it all happens in the blink of an eye.

General search engines make money by selling digital advertisements. Type the words "running shoes" into a general search engine, and sellers of running shoes will compete with one another in a split-second auction to place an advertisement on the results page, which if clicked takes the user directly to the seller's website. This is a highly effective way of reaching consumers. It is also an incredibly lucrative business. In 2021, advertisers spent more than $150 billion to reach users of general search engines.

For more than 15 years, one general search engine has stood above the rest: Google. The brand is synonymous with search. Once a scrappy start-up founded by two Stanford University students in a rented garage, Google is now one of the world's most valuable companies. Its parent company, Alphabet Inc., today has a market capitalization (the value of its outstanding shares of stock) of more than $2 trillion. Much of that value is due to Google's extremely profitable advertising business.

Google's dominance has gone unchallenged for well over a decade. In 2009, 80% of all search queries in the United States already went through Google. That number has only grown. By 2020, it was nearly 90%, and even higher on mobile devices at almost 95%. The second-place search engine, Microsoft's Bing, sees roughly 6% of all search queries—84% fewer than Google.

Google has not achieved market dominance by happenstance. It has hired thousands of highly skilled engineers, innovated consistently, and made shrewd business decisions. The result is the industry's highest quality search engine, which has earned Google the trust of hundreds of millions of daily users.

But Google also has a major, largely unseen advantage over its rivals: default distribution. Most users access a general search engine through a browser (like Apple's Safari) or a search widget that comes preloaded on a mobile device. Those search access points are preset with a "default" search engine. The default is extremely valuable real estate. Because many users simply stick to searching with the default, Google receives billions of queries every day through those access points. Google derives extraordinary volumes of user data from such searches. It then uses that information to improve search quality. Google so values such data that, absent a user-initiated change, it stores 18 months-worth of a user's search history and activity.

The distribution agreements benefit Google in another important way. More users mean more advertisers, and more advertisers mean more revenues. As queries on Google have grown, so too has the amount it earns in advertising dollars. In 2014, Google booked nearly $47 billion in advertising revenue. By 2021, that number had increased more than three-fold to over $146 billion. Bing, by comparison, generated only a fraction of that amount—less than $12 billion in 2022.

For years, Google has secured default placements through distribution contracts. It has entered into such agreements with browser developers, mobile device manufacturers, and wireless carriers. These partners agree to install Google as the search engine that is delivered to the user right out of the box at key search access points.

Google pays huge sums to secure these preloaded defaults. Usually, the amount is calculated as a percentage of the advertising revenue that Google generates from queries run through the default search access points. This is known as "revenue share." In 2021, those payments totaled more than $26 billion. That is nearly four times more than all of Google's other search-specific costs combined. In exchange for revenue share, Google not only receives default placement at the key search access points, but its partners also agree not to preload any other general search engine on the device. Thus, most devices in the United States come preloaded exclusively with Google. These distribution deals have forced Google's rivals to find other ways to reach users.

Google's dominance eventually attracted the attention of antitrust enforcers—the U.S. Department of Justice and nearly every state's Attorney General. They homed in on Google's distribution agreements and in late 2020 filed two separate lawsuits alleging that the agreements and certain other conduct violate Section 2 of the Sherman Act. According to their complaints, Google has unlawfully used the distribution agreements to thwart competition and maintain its monopoly in the market for general search services and in various online advertising markets.

The proceedings that followed have been remarkable. Discovery began in December 2020 and concluded in March 2023. Millions of pages exchanged hands, Google produced petabytes of data, and the parties deposed dozens of witnesses, including high-ranking executives at some of the world's largest technology companies. The court held a nine-week bench trial starting in September 2023. It heard from dozens of live witnesses, including multiple experts, and admitted over 3,500 exhibits. After receiving extensive post-trial submissions, the court held closing arguments over two days in early May 2024. The lawyering has been first rate throughout.

After having carefully considered and weighed the witness testimony and evidence, the court reaches the following conclusion: Google is a monopolist, and it has acted as one to maintain its monopoly. It has violated Section 2 of the Sherman Act.

Specifically, the court holds that (1) there are relevant product markets for general search services and general search text ads; (2) Google has monopoly power in those markets; (3) Google's distribution agreements are exclusive and have anticompetitive effects; and (4) Google has not offered valid procompetitive justifications for those agreements. Importantly, the court also finds that Google has exercised its monopoly power by charging supracompetitive prices for general search text ads. That conduct has allowed Google to earn monopoly profits.

Other determinations favor Google. The court holds that (1) there is a product market for search advertising but that Google lacks monopoly power in that market; (2) there is no product market for general search advertising; and (3) Google is not liable for its actions involving its advertising platform, SA360. The court also declines to sanction Google under Federal Rule of Civil Procedure 37(e) for its failure to preserve its employees' chat messages.

This decision is organized as follows. The court begins with a brief procedural history. It then sets forth findings of fact. They are followed by the court's conclusions of law regarding the challenged distribution agreements. The court first addresses market definition and monopoly power, then the exclusionary nature of the conduct (including the contracts' exclusivity), and finally the agreements' anticompetitive effects and Google's procompetitive justifications for them. A discussion of the SA360-related conduct follows. The opinion ends with brief sections on anticompetitive intent, as well as Plaintiffs' request for sanctions. The court has included as an Appendix a list of the names and titles of all witnesses whose testimony is cited in the decision.

PROCEDURAL HISTORY

On October 20, 2020, the U.S. Department of Justice, joined by 11 States ("U.S. Plaintiffs"), commenced *United States v. Google*, 20-cv-3010 (APM). *See* Compl., ECF No. 1. Pursuant to authority conferred by 15 U.S.C. § 4, U.S. Plaintiffs alleged that Google had violated Section 2 of the Sherman Act by unlawfully maintaining its monopoly in three product markets by entering into exclusive agreements to secure default distribution on nearly all desktop and mobile devices in the United States. *See generally* Am. Compl., ECF No. 94. The alleged markets are general search services, search advertising, and general search text advertising. *Id.* ¶¶ 88–107. U.S. Plaintiffs advanced three Section 2 claims, each corresponding to an alleged market. *Id.* ¶¶ 173–193. They sought a finding of liability, an injunction against the challenged conduct, and structural relief necessary to cure any resulting anticompetitive effects. *Id.* ¶ 194.

On December 17, 2020, 38 States ("Plaintiff States") joined together to bring *State of Colorado v. Google*, 20-cv-3715 (APM) [hereinafter *Colorado v. Google* Docket]. They filed suit pursuant to Section 16 of the Clayton Act, 15 U.S.C. § 26, in their sovereign or quasi-sovereign capacities as *parens patriae* on behalf of the citizens, general welfare, and economy of each of their states. The *Colorado* complaint adopted the allegations in the U.S. Plaintiffs' complaint but supplemented it in three ways. Compl., *Colorado v. Google* Docket, ECF No. 3 [hereinafter *Colorado* Compl.]. First, Plaintiff States alleged a third advertiser-side market for general search advertising but not one, as U.S. Plaintiffs had, for search advertising. *Id.* ¶¶ 56 n.3, 82–89. Second, they asserted exclusionary conduct by Google that targeted specialized vertical providers, or SVPs. *Id.* ¶¶ 168–189. Third, Plaintiff States claimed that Google had engaged in further exclusionary conduct by using its proprietary advertising platform, SA360, to harm competition

in all proposed markets. *Id.* ¶¶ 144–167. Plaintiff States similarly sought declaratory and injunctive relief. *Id.* ¶ 233.

On January 7, 2021, upon Plaintiff States' motion, the court consolidated the two cases for pretrial purposes, including discovery. Order, *Colorado v. Google* Docket, ECF No. 67. The court subsequently consolidated the cases for trial as well. *See* Status Conf. Tr., ECF No. 609, at 10–14. The parties also jointly asked to bifurcate the liability and remedies phases, and the court agreed to do so. *See* Order, ECF No. 264.

Discovery closed on February 23, 2023. Soon after, U.S. Plaintiffs moved for sanctions under Rule 37(e) for Google's failure to preserve relevant chat messages among its employees. Pls.' Mot. for Sanctions, ECF No. 512. The court deferred ruling on the motion pending the presentation of evidence relevant to that issue at trial. Order, ECF No. 610, at 2.

Google also moved for summary judgment in both cases. *See* ECF Nos. 451, 452. The court granted in part and denied in part Google's motions. It entered judgment for Google as to U.S. Plaintiffs' claims related to Android Compatibility Commitments and Anti-Fragmentation Agreements, Google's voice-activated assistant and other "Internet-of-Things" devices, and the Android Open-Source Project. *See United States v. Google LLC*, 687 F. Supp. 3d 48, 78–84, 85–87 (D.D.C. 2023). It also entered judgment in favor of Google on Plaintiff States' theory that Google's targeting of SVPs caused anticompetitive effects in the proposed markets. *Id.* at 78–83. The court permitted the remaining claims to proceed to trial.

Trial commenced on September 12, 2023. Both sides presented exhaustive evidence in support of their various claims and defenses. Dozens of witnesses, including numerous Google employees, third-party witnesses, and several experts, testified live and were subject to lengthy

cross-examination. The parties entered thousands of exhibits and designated certain deposition testimony into the trial record. Trial concluded just over nine weeks later on November 16, 2023.

Following trial, each group of Plaintiffs and Google filed separate post-trial briefs, as well as affirmative and responsive proposed findings of fact and conclusions of law. Those submissions ran into the thousands of pages. Finally, the court held two days of closing arguments on May 2 and 3, 2024.

FINDINGS OF FACT

I. PARTIES AND RELEVANT NONPARTIES

A. Parties

1. Plaintiff United States of America, along with Plaintiffs Arkansas, California, Florida, Georgia, Indiana, Kentucky, Louisiana, Michigan, Mississippi, Missouri, Montana, South Carolina, Texas, and Wisconsin—**U.S. Plaintiffs**—filed the lawsuit captioned *United States v. Google*, 20-cv-3010 (APM). *See* Am. Compl. at 2–3.

2. Separately, Plaintiffs Colorado, Nebraska, Arizona, Iowa, New York, North Carolina, Tennessee, Utah, Alaska, Connecticut, Delaware, District of Columbia, Guam, Hawaii, Idaho, Illinois, Kansas, Maine, Maryland, Massachusetts, Minnesota, Nevada, New Hampshire, New Jersey, New Mexico, North Dakota, Ohio, Oklahoma, Oregon, Pennsylvania, Puerto Rico, Rhode Island, South Dakota, Vermont, Virginia, Washington, West Virginia, and Wyoming—**Plaintiff States**—filed the lawsuit captioned *State of Colorado v. Google*, 20-cv-3715 (APM). *See Colorado* Compl.

3. Alphabet Inc. is the California-based parent company of a collection of businesses, the largest of which is Defendant Google LLC (**Google**). UPX8085 at 851.[1] Google was founded

[1] This opinion uses the last three digits of Bates numbers on an exhibit to cite the specific pages that support a finding of fact.

in 1998 by two students from Stanford University, Larry Page and Sergey Brin, who left school to create Google, which is a general search engine (GSE). Trial Tr. at 7292:21–7293:1 (Raghavan) [hereinafter Tr.]. A GSE is software that produces links to websites and other relevant information in response to a user query. *See infra* Part II. What started in a rented garage is today one of the world's largest companies. The Chief Executive Officer (CEO) of Alphabet and Google is Sundar Pichai. Tr. at 7638:2-12 (Pichai).

4. Although Google began as a GSE, today its core services include a suite of applications widely used on mobile and desktop devices, including Gmail, Google Drive, Google Maps, Google Photos, Google Play, and YouTube. *Id.* at 7717:2-12 (Pichai); UPX8085 at 852.

5. In 2008, Google developed **Android**, an open-source operating system for mobile devices. *See* Tr. at 7652:1–7653:11 (Pichai). An open-source system allows third-party developers to create new smart devices and technologies by customizing the Android system to the device or technology. *See id.* at 7653:2-3 (Pichai) ("[Y]ou can just take the open source project and do whatever you want with it without ever talking to Google"); *id.* at 9414:25–9415:3 (Rosenberg) ("Being open source, [Android is] customizable. It [i]s something that someone could take with its underlying capabilities and then build on top of and add capabilities to."). Today, hundreds of millions of mobile devices in the United States run on the Android operating system. UPX639 at 266.

6. Also in 2008, Google launched **Chrome**, a web browser. Tr. at 7646:5-7 (Pichai). A web browser is software that allows users to access websites on the internet, among other things. *See* M. Baker Dep. Tr. at 23:1–27:8. Chrome was designed to increase the speed and seamlessness of web navigation by users. Tr. at 7649:11–7650:2 (Pichai). "Chromium is the underlying engine

which powers Chrome," and it is fully open source, like Android. *Id.* at 7648:21–7649:5 (Pichai). Google is the default search engine on Chrome. *Id.* at 7650:5-9 (Pichai).

7. Google also acquired an online advertising platform, DoubleClick, in 2008, which it developed into what today is known as **SA360**. *Id.* at 1235:5-12 (Dischler); PSX1109. SA360 is a search engine marketing tool, which allows advertisers to purchase digital advertisements across multiple platforms. Tr. at 1234:2–1235:4 (Dischler); *see also infra* Section V.G.

8. In 2022, Google reported Search+ revenues over $162 billion. UPX8085 at 879, 899 (including "other Google owned and operated properties like Gmail, Google Maps, and Google Play"). Between 2014 and 2021, Google's Search+ revenues more than tripled, with gross margins ranging from 76–82% annually. *See* UPX7002.A. The vast majority of Alphabet's revenues (nearly 80%) come from digital advertisements, and historically the largest component has been ads displayed on Google's search engine results page. *See* UPX8085 at 878–89; UPX342 at 824 (attributing approximately 66% of the "company's revenue and $ growth for 10+ years" to search advertising).

B. Key Third Parties

9. **Apple Inc.** is a California-based company that "designs, manufactures[,] and markets smartphones, personal computers, tablets, wearables[,] and accessories, and sells a variety of related services." UPX8105 at 172, 175. Those products include the iPhone, iPad, and Mac personal computers (PCs). *Id.* at 175. Each of these devices runs on an Apple-developed, proprietary operating system: iOS for iPhones, iPadOS for iPads, and macOS for Mac computers. *Id.* Unlike Android, Apple's operating systems are not open source. *See* Tr. at 9841:25–9842:5 (Murphy). Apple's products all come preloaded with its proprietary web browser, Safari. *Id.* at

632:9-10 (Rangel). In 2022, Apple's market capitalization was at least $2.8 trillion. UPX8105 at 173.

10. **Microsoft Corporation** is a Washington-based company whose products include an operating system called Windows, a web browser called Edge, and various devices, including PCs and tablets. UPX8094 at 517, 521, 530–31. In 1998, Microsoft licensed a third-party GSE, MSN Search, for use on its devices. Tr. at 3545:11-21 (Nadella). In 2005, Microsoft created its own GSE, which was then known as Live Search. *Id.* at 3547:3-24 (Nadella). In 2009, Microsoft launched **Bing**, a GSE. *Id.* at 3548:4-5 (Nadella). Microsoft has invested nearly $100 billion into search over the past two decades. *Id.* at 3510:3-7, 18-21 (Nadella). Bing's search and news advertising revenue totaled $11.6 billion in 2022. *See* UPX8094 at 612. The CEO of Microsoft is Satya Nadella. Tr. at 3487:2-6 (Nadella). Microsoft's revenues in 2022 were over $198 billion, with a market capitalization of $2.5 trillion. UPX8094 at 559, 517.

11. **Mozilla Corporation** is a California-based company that developed an open-source web browser called Firefox for both desktop and mobile devices. JX31 at 612, 633. Today, Mozilla's share in the desktop browser market is about 10% and negligible in the mobile market. M. Baker Dep. Tr. at 127:24–128:8, 134:9-20. In 2018, Mozilla generated $435 million in revenues. *Id.* at 285:12-16 (discussing UPX979 at 414).

12. **DuckDuckGo (DDG)** is Pennsylvania-based web services company founded in 2008. Tr. at 1937:4-7 (Weinberg). It offers a product that is an integrated browser and GSE. *Id.* at 1962:6-12, 1963:3-16 (Weinberg). Gabriel Weinberg is the founder and CEO of DDG. *Id.* at 1937:2-3 (Weinberg). DDG does not produce its own search results or search advertisements. It syndicates both from Microsoft. *Id.* at 3510:8-11, 3520:13-22 (Nadella). DDG attempts to

differentiate itself from other GSEs through a focus on user privacy. *E.g.*, *id.* at 1937:14-20, 2150:13-18 (Weinberg).

13. **Yahoo** is a California-based provider of general search services and was an early market leader in general search. UPX1053 at 121; Ramalingam Dep. Tr. at 23:2-12. In 1998, the year that Google was founded, Yahoo already had hundreds of millions of users. UPX1053 at 121. By 2009, however, Yahoo had stopped crawling the web and producing its own search results. Instead, it reached a data-sharing and syndication agreement with Microsoft, which provided that the two companies would combine their search engine user data (primarily to compete with Google) and, going forward, Yahoo's search results would be delivered by Bing. *See* DX271; Tr. at 3520:13–3522:9 (Nadella). Yahoo also has popular subject-specific, or "vertical," products, such as Yahoo Sports and Yahoo Finance. Ramalingam Dep. Tr. at 24:14–25:11.

14. **Neeva** was a California-based company incorporated in 2017 that introduced a new GSE in 2019. Tr. at 3670:1-5, 3670:24–3671:1 (Ramaswamy). Neeva was founded by Dr. Sridhar Ramaswamy, a veteran Google Search executive. *Id.* at 3667:3–3669:14 (Ramaswamy). One of Neeva's distinguishing features was that it was a subscription-based service that did not serve advertisements. *Id.* at 3675:22–3679:16 (Ramaswamy). Although Neeva initially licensed Bing's search infrastructure to respond to all queries, by 2022 Neeva responded to about 60% of queries using its own systems, relying on Bing for the remainder. *Id.* at 3739:14-16, 3776:14-21 (Ramaswamy). In May 2023, Neeva shut down and was acquired by Snowflake Inc., an enterprise data company. *Id.* at 3675:1-6 (Ramaswamy). It no longer exists as a GSE. *Id.* at 3675:5-19 (Ramaswamy).

15. **Branch** is a California-based software company that was founded by Stanford graduate students in 2013. *Id.* at 2892:7-24 (Austin). Branch created a search engine for

applications using "deep linking" technology, which allows users to search across pages of mobile applications on a particular device and navigate to relevant application results. *Id.* at 2893:18–2894:18, 2897:3-15 (Austin). This "app search engine" required "work[ing] individually with every app company [to] get them to send [] the actual pages inside of the app," which entailed "build[ing] the one-on-one relationship with the app [and] hav[ing] them develop, write custom code." *Id.* at 2898:2-9 (Austin). Unlike a GSE, Branch's product does not index the web and (in its presently deployed version) does not deliver web results. *Id.* at 2957:3-15, 2956:16-24 (Austin); *see also infra* Section VI.B.2.d.

16. **Samsung Electronics Co. Ltd.** is a Korea-based original equipment manufacturer (OEM) of smartphones and other mobile devices that run on the Android platform. *See* UPX639 at 266; Baxter Dep. Tr. at 34:14. Samsung devices "represent the primary competitor to the iPhone in key monetizing regions, such as the US[.]" UPX639 at 266. Samsung develops mobile applications that it preloads onto its devices, including a browser known as S Browser and an app store called the Galaxy Store. Baxter Dep. Tr. at 83:10-24, 91:4-7. Samsung also invests in novel products through its innovation arm, Samsung Next. Tr. at 4485:3-12, 4485:22–4486:14 (Chang).

17. **Motorola Mobility LLC** is an Illinois-based OEM of smartphones that run on the Android platform. JX39 at 794. Motorola and Samsung together manufacture the majority of Android devices in the United States. Tr. at 775:2-5 (Kolotouros). Google acquired Motorola but later sold it to Lenovo Group Ltd. Christensen Dep. Tr. at 15:12-14, 142:12-18.

18. **AT&T Mobility LLC** is a Georgia-based mobile carrier that provides wireless services that connect mobile devices to cellular networks. JX91 at 742. AT&T also sells devices directly to consumers. Ezell Dep. Tr. at 28:4-12. Roughly 30% of the smartphones that it distributes are Android devices. *Id.* at 29:8-25. The other 70% are Apple devices. *Id.*

19. **T-Mobile US, Inc.** is a Washington-based mobile carrier that provides cellular services and sells mobile devices directly to consumers. JX95 at 687–88; Tr. at 9313:24-25 (McCallister). Approximately half of the phones sold by T-Mobile run on Android, and the other half are Apple devices. Giard Dep. Tr. at 23:16–24:7.

20. Cellco Partnership, doing business as **Verizon Wireless**, is a New Jersey-based mobile carrier that provides cellular services and sells mobile devices directly to consumers. JX93 at 487–88; Tr. at 9313:24-25 (McCallister). It distributes roughly twice as many Apple devices (70%) as Android devices (30%). Tr. at 1102:21-23 (Higgins).

II. GENERAL SEARCH ENGINES

A. Overview

21. Google, Bing, Yahoo, DDG, Ecosia, and Brave are GSEs. *See, e.g.*, *id.* at 2168:1-4 (Giannandrea); *id.* at 1031:2-10 (Higgins); *id.* at 1942:11-17 (Weinberg); *id.* at 8201:23-24 (Reid). There is "relatively limited [user] overlap between the general search engines." *Id.* at 8728:23-24 (Israel).

22. Bing is Google's largest general search competitor today. *Id.* at 8094:8-10 (Raghavan). It is the only rival that crawls the web and generates its own search results. The next two largest search engines, Yahoo and DDG, syndicate their search results from Bing. *See id.* at 3520:13-25 (Nadella).

23. By 2009, 80% of all general search queries, whether entered on a desktop computer or mobile device, flowed through Google. *Id.* at 4762:4-12 (Whinston) (discussing UPXD102 at 48); *e.g.*, UPX472; *see also* Tr. at 203:21–204:5 (Varian) (Google began measuring its search share against other GSEs monthly in 2009). That percentage had increased from 80% to 89.2% by 2020. Tr. at 4761:14–4762:8 (Whinston) (discussing UPXD102 at 47).

24. Google's share of search queries on mobile devices was even higher at 94.9% in 2020. *Id.* at 4762:19–4763:2 (Whinston) (discussing UPXD102 at 49); *see also* UPX476 at 668 (Google's internal share calculation of 98% of mobile GSE queries in 2019). The percentage on desktop devices was 84%. *See* UPX476 at 668.

25. Google's second-place rival, Bing, receives roughly 6% of all search queries. Tr. at 4761:12-14 (Whinston) (discussing UPXD102 at 47). Bing (5.5%), Yahoo (2.2%), DDG (2.1%), and other rivals (0.9%) together see less than 11% of all queries. *Id.* Their numbers are even lower on mobile devices. *Id.* at 4762:19–4763:2 (Whinston) (discussing UPXD102 at 49) (Bing (1.3%), Yahoo (2.1%), DDG (1.5%), and others (0.2%)). Bing's market share has never risen above 12%. *See id.* at 4762:4-12 (Whinston) (discussing UPXD102 at 48).

26. Bing sees more desktop queries than mobile queries because it has greater distribution on Windows desktop devices, where it is the default GSE on the preloaded Edge browser. *See id.* at 3096:14-18 (Tinter).

B. How a GSE Works (Greatly Simplified)

27. "A general search engine is a tool that you use to search the worldwide web using queries." *Id.* at 2167:3-4 (Giannandrea). A GSE attempts to answer all queries by "provid[ing] search results that are relevant to those queries." *Id.* at 8093:10-12 (Raghavan); *id.* at 182:6-8 (Varian). "The primary source of information for Search is the web." UPX194 at 552.

28. The first step in developing a search engine is to crawl the web. *Id.* at 552; Tr. at 1774:20-22 (Lehman); *id.* at 2206:14-15 (Giannandrea) ("[S]tep one [of] building a general search engine would be to take a copy of as much of the web as you can."). GSEs crawl the web using a "crawling bot," which "starts with a list of websites[.]" Tr. at 2206:17-20 (Giannandrea). The bot "crawls the HTML on those websites and then it looks at the links inside of those web pages and

then recursively crawls them." *Id.* And, because websites "are constantly changing and the web is constantly growing," GSEs "constantly recrawl the web to index new content." UPX194 at 552–53.

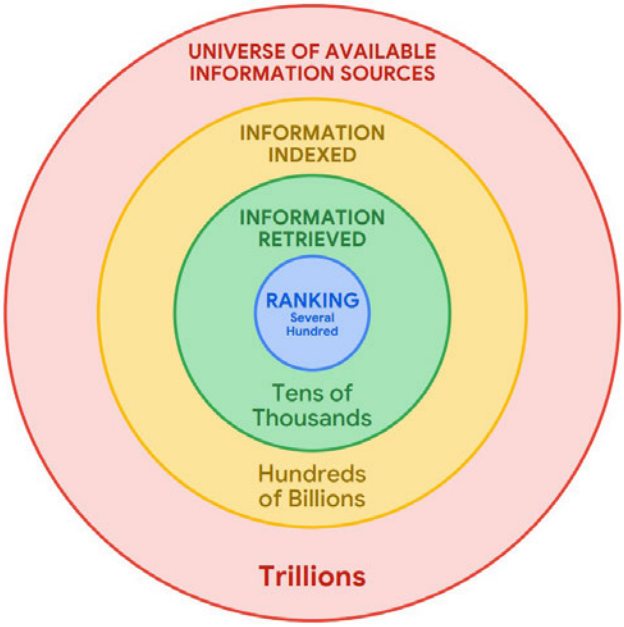
29. The results of the web crawling are organized into an index. An index is "a database essentially of the whole web that's publicly available that can be returned if [a] user asks for it." Tr. at 2656:17-18 (Parakhin). The development of an index is "a crucial piece of the puzzle," because if a site is not in the index, it will not be presented to users in response to a query. *Id.* at 6303:20-25 (Nayak); *id.* at 2210:21 (Giannandrea) ("What you include in the index matters a lot[.]"). Thus, the more sites in an index, the better. *Id.* at 2212:4 (Giannandrea). Today, only Google and Bing create fulsome web search indexes that generate accessible results. DDG indexes portions of the web to create its own search "modules." *Id.* at 1939:2–1941:16 (Weinberg). And Apple maintains an index of about [redacted] billion websites, although it does not presently plan to use that index to offer a results page. *Id.* at 2212:9-14 (Giannandrea); FOF ¶ 302.

30. An index is only useful if the GSE understands what the user is seeking with a query. GSEs "aim to identify spelling errors, annotate the query with synonyms, mark multi-word concepts, generate terms related to the query, and more." UPX213 at 715. Google does this in many ways: through its spelling and synonyms functions, using "query-based salient terms" (QBST) that are likely to show up in a responsive document, and semantic tools, such as query clustering and segmentation. *Id.* at 715–16; *see also* UPX870 at .016–.017.

31. The GSE then must retrieve and rank websites responsive to the query. Common queries can yield a nearly infinite number of potentially responsive sites, so the GSE must include a retrieval system that narrows the volume of responsive links to tens of thousands, as opposed to millions. Tr. at 6331:7-15 (Nayak). The GSE then must rank these several thousand results. It

first must decide which results are worth scoring at a more granular level, and then score those hundreds of sites to determine which top 10 or so should be surfaced to the user. *Id.* at 6331:13–6332:11 (Nayak); *infra* Section II.G.

32. The above-described culling and sorting process by which a GSE produces search results is illustrated below:



DXD17 at 2.

C. Types of Queries

33. A GSE can supply information from a broad variety of sources, covering nearly any topic. Tr. at 8708:16-20 (Israel) (agreeing that GSEs "can handle virtually any type of query"). Thus, it is "the first place that you can turn to," and "a place that you go to for the vast majority of your information needs." *Id.* at 3670:6-18 (Ramaswamy); *see also id.* at 6511:11-23 (Whinston) (same); *id.* at 7027:23-25 (J. Baker) ("[A] general search user can get satisfactory responses to multiple queries from multiple sources, all without switching sites."); *id.* at 10471:17-25 (Oard)

("[M]ental process is costly[] and . . . people may just not know about where things are."); *cf. id.* at 8717:17-18 (Israel) ("If I don't know the best source, I may have to try various ones.").

34. Google classifies its queries by subject matter, and it has developed more than two dozen "level-one" classifications. *Id.* at 7029:2-16 (J. Baker) (discussing PSDX11 at 17). Users tend to use a GSE for a short period of time to search for a particular topic and then allow time to pass before using a GSE to search for a different topic. Put differently, users do not typically search multiple different subject matters during an unbroken time period. *Id.* at 8419:9-15 (Israel) (discussing DXD29 at 25) (78% of users searched within only one vertical in a short period). Yet, if viewed over a longer period, users frequently turn to GSEs to search for a broad variety of topics. *See id.* at 7029:17–7031:11 (J. Baker) (discussing PSXD11 at 19) (showing based on Google sessions data from 2019 and 2021 that nearly 65% of "sessions," defined in the study as a 24-hour period, involved users searching in more than one classified segment).

35. Many users begin their online information gathering journeys on GSEs. An analysis by U.S. Plaintiffs' expert Dr. Michael Whinston found that 77% of search sessions on Windows desktop devices began on GSEs. *Id.* at 4614:12-24 (Whinston). That 77% figure is arguably lower on mobile devices, on which users are more likely to start searches directly within an application instead of a GSE. *See id.* at 5875:19–5876:9 (Whinston).

36. There are two general types of queries on GSEs: noncommercial and commercial.

37. A noncommercial query is one in which the user seeks to retrieve information that the GSE does not attempt to monetize by delivering a search advertisement. 80% of Google's queries are noncommercial in nature. *Id.* at 8396:16–8398:17 (Israel); UPX10 at 053 n.6.

38. Commercial queries, as the name implies, are queries that the GSE perceives are an expression of commercial intent by the user and constitute the remaining 20% of Google's queries.

Tr. at 8396:16–8398:17 (Israel); UPX10 at 053 n.6. Typically, such a query seeks information on a product or a service. GSEs often serve advertisements on a search engine results page in response to a commercial query. *See infra* Section V.A.1. Like Google, only about 20–30% of Bing's queries are commercial and show ads. Tr. at 3645:13–3646:2 (Nadella).

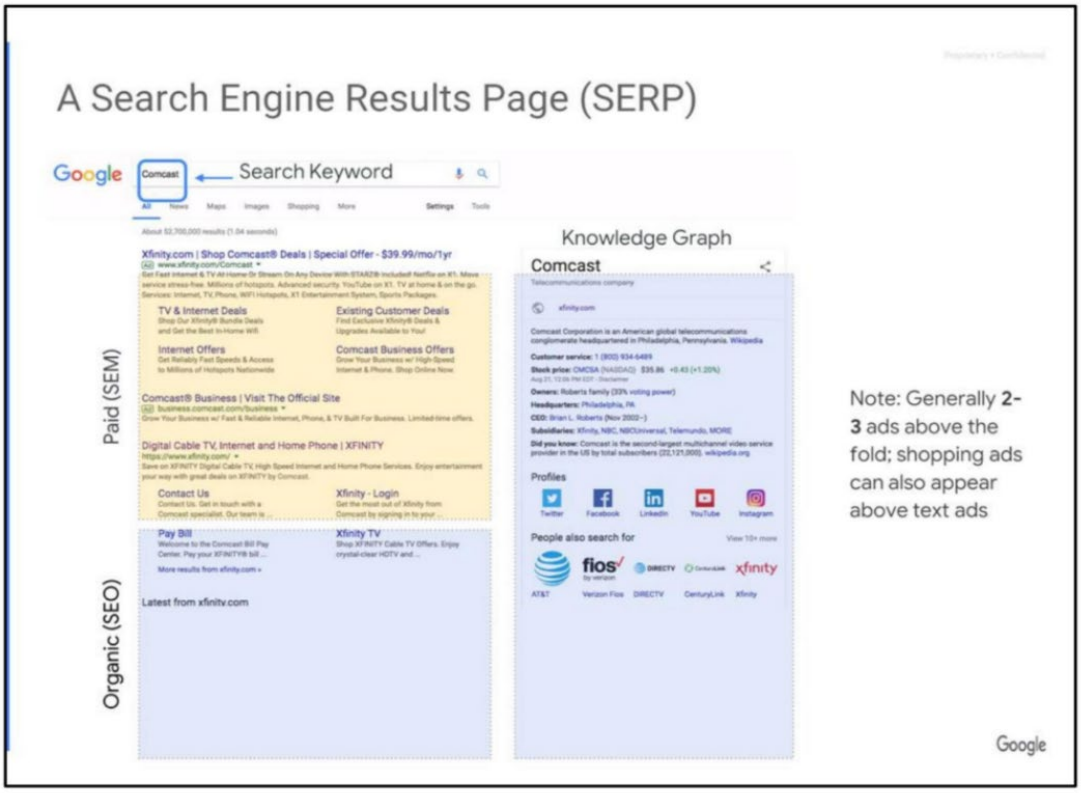
39. Navigational queries, which can be either commercial or non-commercial, are a type of query that reflects a user's intent to navigate directly to a particular website. *Id.* at 185:11-19 (Varian). GSEs may or may not serve ads on a navigational query. An example of a navigational query is "Amazon," which may express the user's intent to go to Amazon.com. *See id.* at 8721:12-13 (Israel) ("[O]ne use of a general search engine[] is as this vehicle to take me to other sites."). Users often enter navigational queries. In fact, at a given time, Google's top five queries by query volume are navigational queries, UPX342 at 859, and nearly 12% of all Google queries are navigational queries, Tr. at 8748:22–8749:1 (Israel) (calculation reflected in Whinston Expert Report at 64); *id.* at 8748:25–8749:1 (Israel) (the volume of navigational queries is "significant"). Navigational queries are unique to GSEs, because only a GSE's results page supplies a user with organic links used to navigate to another website. *See id.* at 4616:23-25 (Whinston) (specialized vertical providers are "not sending you off to other sites" because "they don't have a broad index of the web"); *see infra* Section IV.A.

40. The number of general search queries has grown dramatically over the last decade, especially on mobile devices. *See* Tr. at 8442:17–8443:2 (Israel) (discussing DXD29 at 45) ("[O]utput is more than double over this 10-year time period."); *id.* at 7248:4-10 (J. Baker) (discussing PSXD12).

D. Search Engine Results Page

41. GSEs produce information responsive to a query on a search engine results page, or SERP. The SERP "provid[es] links to websites drawn from a broad index of the web as well as provid[es] additional information[.]" *Id.* at 4610:21-22 (Whinston); *id.* at 7026:20-22 (J. Baker).

42. Most SERPs contain some mixture of advertisements, organic links, and vertical offerings. A sample SERP is illustrated below.



UPX1 at 533.

43. Organic links, or "blue links," are unpaid search results that allow a user to navigate directly to a website. Tr. at 2221:15-19 (Giannandrea); *id.* at 6509:25–6510:1 (Nayak). A GSE determines which links to present by sorting through indexed webpages and presenting relevant results. *See* UPX8104 at 165; *see also supra* Section II.B; *infra* Sections II.G & II.H.

44. Paid advertisements are typically generated in response to a commercial query and usually appear at the top of a SERP. *See* UPX1 at 533. Multiple types of advertisements can appear on a SERP, but the two primary ones are general search text ads (which resemble organic results but are labeled "sponsored" on Google) and shopping ads (which typically consist of a product photograph, vendor identity, and price information). *See infra* Section V.A.1.

45. A vertical offering is a category of specialized information that is accessible to users without leaving the SERP. Tr. at 2336:14-16 (Giannandrea); *id.* at 6509:7-21 (Nayak). Examples of verticals include information about flights, hotels, and restaurants. Such information is usually acquired from third parties and is referred to as "structured data." *Id.* at 8224:18–8225:6 (Reid). Structured data can come from several sources: specialized vertical providers (like online travel sites), users, merchants, or GSE employees in the field. *Id.* Much of "th[is] information is not even on the web." *Id.* at 8224:24-25 (Reid). Another example of structured data is a "knowledge graph," which is a database containing useful information about people, places, and things, as well as the connections among them. *See* Moxley 30(b)(1) Dep. Tr. at 17:17-20; UPX1 at 533.

46. GSEs enter into data-sharing agreements with partners (usually specialized vertical providers) to obtain structured data for use in verticals. Tr. at 9148:2-5 (Holden) ("[W]e started to gather what we would call structured data, where you need to enter into relationships with partners to gather this data that's not generally available on the web. It can't be crawled."). These agreements can take various forms. The GSE might offer traffic to the provider in exchange for information (i.e., data-for-traffic agreements), pay the provider revenue share, or simply compensate the provider for the information. *Id.* at 6181:7-18 (Barrett-Bowen).

47. As of 2020, Microsoft has partnered with more than 100 providers to obtain structured data, and those partners include information sources like Fandango, Glassdoor, IMDb,